Highway Engineering Problem Session-2

Question 1: A two-way-two-lane highway started to operate in 1998. In the first year 1,500,000 vehicles, and in the second year 1,590,000 vehicles used the highway. For each year the heavy vehicle rate is given as 20%.

a) Check if it achieves to reach 20 years of service life with service level C as predicted in the design stage.
b) If the right of way is sufficient, in order to achieve the predicted life span, when should new lanes be added?

Given: Necessary arrangments have been performed within the service volume table below. The highway is required to provide service level C at the end of its life span. K=1/6; constant traffic increase; 1 light vehicle = 1 bo, 1 heavy vehicle= 3 bo, (r) pce: passenger car equivalent

Service Level	Service Volume (bo/h/ln)		
	2 way 2 lanes	3 lanes	2x2 lanes
A	100	200	400
B	200	500	700
C	600	750	1000
D	800	1000	1500
E	1000	1400	1900

1998 : 1500000 vehicles/year
1500000 x (0.8x1 + 0.2x3) = 2100000 pce/year
AADT: annual average daily traffic (yıllık ortalama günlük trafik)
AADT = 2100000 pce/year / 365 = 5753.42 pce/day
DHT: design hourly traffic (projeasatlik trafiği)
DHT = AADT . K

Solution:

AADT . K; light vehicle; heavy vehicle

a) $$T_{1998} = \frac{1{,}500{,}000 * (0.80*1 + 0.20*3)}{365*6} = 959 \text{ bo/h (pce/h)}$$

$$T_{1999} = \frac{1{,}590{,}000 * (0.80*1 + 0.20*3)}{365*6} = 1017 \text{ bo/h}$$

$$T_n = T_o * (1+r)^n \qquad 1017 = 959 * (1+r)^1 \qquad r = \%6$$

1 year difference

For Service Level C:

increase

$$2*600 = 959*(1+0.06)^n \qquad \ln\frac{1200}{959} = n*\ln(1.06) \qquad n=3.85 \text{ years}$$

pce/h ← ln x pce/h/ln pce/h

3.85 years < 20 years therefore life span prediction fails.

b) 1 lane should be added after 3.85 years (it will have 3 lanes)

pce/h

$$3*750 = 959*(1+0.06)^n \qquad n=14.64 \text{ years}$$

lane pce/hour/lane

So after n=14.64 years one more lane should be added.

In 1998 the highway starts as 2 lanes

In 1998+3.85 the highway gets a 3rd lane

In 1998+14.64 the highway gets a 4th lane and becomes 2x2

Question 2: The number of vehicles using a two-way-two-lane highway in 1987 and 2002 are at the limits of service levels B and C, respectively.

a) How many years after 1987 does it take for this highway to reach the limit of service level D?

b) Compare the traffic densities of the complete platform for 1987 and 2002.

c) For how many years after 1985 is it possible to keep the free flow speed if the number of vehicles using the highway is 300,000 veh/year for 1985?

Given: Necessary arrangments have been performed within the service volume table below. Ignore the ratio of heavy vehicles. Increase ratio of traffic is constant. K=1/6

Service Level	Service Volume (veh/h/ln)	Operational Speed (km/h)
A	100	100
B	200	85
C	550	65
D	800	50

a)

B service level in 1987

C service level in 2002

$T_{2002} = T_{1985} * (1+r)^{15}$ $\quad 550 = 200 * (1+r)^{15}$ $\quad r = \%\ 7$

For Servicel level D ;

$800 = 200 * (1+0.07)^{n}$ $\quad \ln\frac{800}{200} = n * \ln(1.07)$ $\quad$ n=20.5 years

The highway reaches the limit of service level D 20.5 years after 1987.

b)

q=k*u (Volume=Density*Speed)

$k_{1987} = \frac{200 * 2}{85}$ = 4.71 veh/km $\quad k_{2002} = \frac{550 * 2}{65}$ = 16.92 veh/km

c)

$T_{1985} = \frac{300{,}000}{365 * 6}$ = 137 veh/h

Free flow only possible at Service level A !!!

For service level A;

$2 * 100 = 137 * (1+0.07)^{n}$ $\quad \ln\frac{200}{137} = n * \ln(1.07)$ $\quad$ n=5.59 years

It is possible to keep free flow for 5.59 years.

Question 3: On rolling terrain, there is a double-way road that consists two lanes in each direction. 2600 vehicles use this road in each direction hourly including 18% truck and 5% bus type. The lane width is 3.30 m and the shoulder width is 2 m. The max allowable speed is 80 km/h. Are the number of lanes sufficient to serve at level D?

Service level Data for multiple lane roads

Service Level	Density (veh/km/ln)	Vp=110 km/h			Vp=100 km/h			Vp=80 km/h		
		Vort. (km/h)	Q/C	qm (veh/h/ln)	Vort. (km/h)	Q/C	qm (veh/h/ln)	Vort. (km/sa)	Q/C	qm (veh/h/ln)
A	≤7,5	≥91	0.36	700	≥80	0.33	650	-	-	-
B	≤12,5	≥85	0.54	1,100	≥77	0.50	1,000	≥67	0.45	850
C	≤18,8	≥80	0.71	1,400	≥70	0.65	1,300	≥62	0.60	1,150
D	≤26,3	≥64	0.87	1,750	≥64	0.80	1,600	≥56	0.76	1,450
E	≤41,9	≥48	1.00	2,000	≥48	1.00	2,000	≥45	1.00	1,900
F	>41,9	<48	d	d	<48	d	d	<45	d	d

d: very inconsistent

Factors used for narrowlane and banquets for multiple lane roads (f_w)

Distance to obstacle (m)*	Factors (f_w)							
	obstacle on one side				obstacle on both sides			
	Lane width (m)							
	3.7	3.3	3.0	2.7	3.7	3.3	3.0	2.7
divided multpiple lane road two lanes each direction								
≥2,0	1.00	0.97	0.91	0.81	1.00	0.97	0.91	0.81
1.0	0.98	0.95	0.89	0.80	0.97	0.94	0.88	0.78
0	0.90	0.87	0.82	0.73	0.81	0.79	0.74	0.66
divided multpiple lane road three lanes each direction								
≥2,0	1.00	0.96	0.89	0.78	1.00	0.96	0.89	0.78
1.0	0.98	0.94	0.88	0.77	0.97	0.93	0.86	0.76
0	0.94	0.91	0.85	0.74	0.91	0.87	0.81	0.70
undivided multpiple lane road two lanes each direction								
≥2,0	1.00	0.95	0.89	0.77	N	N	N	N
1.0	0.97	0.93	0.87	0.76	N	N	N	N
0	0.88	0.85	0.80	0.70	0.81	0.79	0.74	0.66
undivided multpiple lane road three lanes each direction								
≥2,0	1.00	0.95	0.89	0.77	N	N	N	N
1.0	0.98	0.94	0.88	0.76	N	N	N	N
0	0.94	0.90	0.83	0.72	0.91	0.87	0.81	0.70

* Take average if the obstacles have different distances

N: use factors that obstacle on one side

Vehicle Type	Passenger car equivalents of Bus and Trucks		
	Terrain		
	Straight	Rolling	Mountain
Truck	1.7	4 pce	8
Bus	1.5	3 pce	5

	Factors reflecting environmental features (fe)	
Residantial type	Road type	
	Divided	Undivided
Rural	1	0.95
Suburb	0.9	0.8

Driver Factors (fp)	
Driver Type	(Fp)
Usual Driver	1
Rare Driver	0.75-0.90

Given:

Traffic in one direction : 2600 veh/h
Terrain : Rolling
Truck rate : 18%
Bus rate : 5%
Lane width : 3.30 m
Banquet width : 2 m
Driver Feature : usual driver
Project speed : 80 km/h
Service level : D

$Q_{SFi} = C_j \cdot (Q/C)_i \cdot N \cdot f_w \cdot f_{HV} \cdot f_e \cdot f_p$

Q_{SFi} :For service level (i), service volume in one direction (veh/h).
C_j:If the design speed is 80 km/h; C_j = 1900 veh/h/ln; if the design speed is ≥100 km/h; C_j = 2000 veh/h/ln;
(Q/C)i: For service level (i) volume/capacity ratio
N: Number of lanes for each direction
f_w: Adjustment factor for lane width and shoulder width
f_{HV} : Adjustment factor for heavy vehicle traffic
f_e: Adjustment factor for environmental feature
f_p : Adjustment factor for driver feature

Solution:

Project speed=80 km/h C_j=1900 veh/h/ln

Q/C = 0.76
f_w = 0.95

$$f_{HV} = \frac{1}{[1 + P_T(E_T - 1) + P_B(E_B - 1)]} = \frac{1}{[1 + 0.18\ \ (4-1) + 0.05(3-1)]} = 0.61$$

f_e = (undivided road) = 0.95
f_p = (usual driver) = 1
$Q_{SFi} = C_j \cdot (Q/C)_i \cdot N \cdot f_w \cdot f_{HV} \cdot f_e \cdot f_p$ substitute those into the equation

$2600 = 1900 \cdot 0.76 \cdot N \cdot 0.95 \cdot 0.61 \cdot 0.95 \cdot 1$

$N = 3.27 \rightarrow$ 4 lanes are required. Add 2 more lanes in each direction.